Improving Generalist Policies with Simulation Post-Training

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Abstract: We present an ongoing study of how simulution can be used for post-training generalist robot policies. We consider the setting where a pretrained generalist policy already exhibits partial competence on a target task and investigate a recipe to improve the policy without additional human demonstrations or enforcing task structure beyond sparse rewards. Our work focuses on unpacking the expert learning process, which most sim-to-real works treat as a black box, to simplify state-based expert learning and improve downstream student-teacher distillation. We leverage the following insights to setup a recipe for policy improvement with simulation: *generalist policy rollouts can reduce the burden of exploration in expert learning* and *artifacts of expert learning, particularly the priveleged critic, can improve the distillation step beyond standard supervised finetuning*. We additionally discuss ongoing work on pushing performance, expanding recovery behavior, and policy improvement without needing distillation.

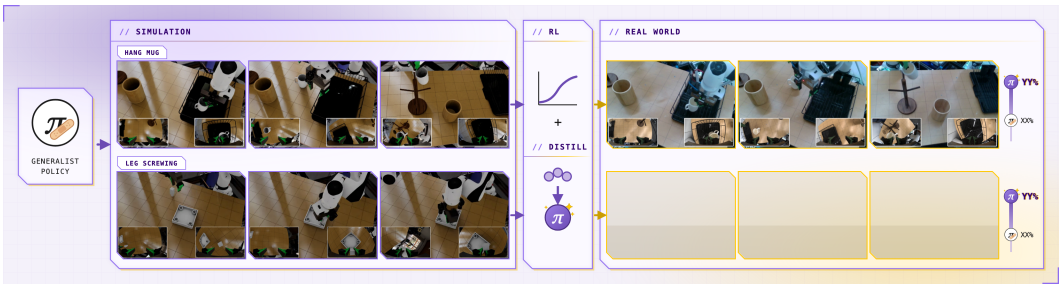
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1 Introduction

Recent years have brought growing interest to robotic foundation models, trained at scale on human teleoperation data [1, 2], video [3], and egocentric data [4, 5]. Whilst broad pretraining has enabled pretrained policies to solve unseen tasks and operate in unseen environments, they often lack the performance and consistency to be qualified to be deployed reliably zero-shot. Additionally, just scaling up teleoperation data yields diminishing returns [6]. This motivates methods of post-training, which build on a pretrained policy and use more targeted experiences to improve its task performance. Recent work has explored methods of policy improvement in offline [7, 8] and online [9, 10] settings. However, relying on real world data can be slow to collect, limited in throughput by hardware and labor costs.

Simulation offers a way to gather this experience at scale, with additional benefits of massively parallel rollouts, access to privileged state, and reset to task or recovery states. In the past, many works have explored using simulation for data generation and training policies for sim-to-real [11, 12]. Existing works in sim-to-real often adopt the paradigm of training a state based expert in simulation, then distilling to a vision-based policy for transfer to the real world [13, 14]. However, these approaches typically treat the privileged expert as a black box, separating the process of expert learning from downstream student distillation. This separation overlooks two opportunities. First, training a high-performing expert from scratch can be challenging. Long-horizon tasks with sparse rewards often requires carefully designed reward shaping or task-specific curricula [15]. Yet, *when a pretrained generalist policy is already available, its rollouts, despite being suboptimal, can provide useful priors for expert exploration*. Second, expert training produces more than just an optimized policy. In particular, most works use some form of reinforcement learning which produce critics capturing the long-term consequences of actions. *These expert-learned value functions can be utilized to improve distillation rather than discarding them.*

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2 Method

Overview Our goal is to take our subpar pretrained policy and turn it into a master of a target task. We assume access to a simulated environment mirroring the real world setting, which can be achieved with hand-crafting or real-to-sim methods [16]. To start, we adopt a common paradigm in sim-to-real of training a privileged expert in simulation [13, 14, 15]. We bootstrap with knowledge from our pretrained policy to guide exploration under long horizon sparse reward settings. Massive GPU parallelism allows us to train an expert with high success rates, which we then use to distill into the pretrained policy with a slight change in distillation objective using the expert's critic.

2.1 State-based Expert Learning

We start with a task where our pretrained policy has non-zero performance. The pretrained policy is rolled out zero-shot in simulation to collect reasonable but imperfect rollouts, yielding transitions of $\mathcal{R}_{policy} = \{(s_t, a_t, r_t, s_{t+1})\}$. The state visitation distribution from the pretrained policy serves as a useful prior to guide exploration for state-based RL, which is especially helpful under long horizon sparse reward settings. We use the FastSAC [17] implementation for our expert learning. To leverage the state visitation prior, we aggregate states from our rollouts (including success and failure) to a reset state buffer, $\mathcal{S}_{rollout} = \{s_t \mid (s_t, a_t, r_t, s_{t+1}) \sim \mathcal{R}_{policy}\}$. During environment resets, we sample from a weighted mixture of rollout states and task initial condition ($\mathcal{S}_{IC}$) state buffers, $s_0 \sim \lambda\mathcal{S}_{rollout} + (1-\lambda)\mathcal{S}_{IC}, \lambda \in [0, 1]$, to achieve the effect of guided exploration without any environment specific reward shaping or structure. Rollouts also come with an action prior that can be a useful signal during expert learning. We instantiate this by maintaining an RLPD-style [18] offline replay buffer $\mathcal{R}_{offline} = \{(s_t, a_t, r_t, s_{t+1}) \in \mathcal{R}_{policy} \mid \text{outcome} = \text{success}\}$, which we sample from in our actor-critic updates with the weighted mixture: $(s_t, a_t, r_t, s_{t+1}) \sim \alpha\mathcal{R}_{offline} + (1-\alpha)\mathcal{R}_{online}, \alpha \in [0, 1]$.

2.2 Base Policy Improvement

Having trained a high-performing privileged expert, we aim to distill its behavior back into the pretrained visual policy. Supervised fine-tuning (SFT) on expert demonstrations suffers from covariate shift, as errors made by the student can lead to states outside the expert's training distribution. DAgger [19] addresses this by iteratively collecting student-visited states and querying the expert for corrective actions.

SFT and DAgger. We first warm-start the student with SFT on expert trajectories, followed by a windowed variant of DAgger. The student π_θ predicts action chunks of length H, executed for $K \leq H$ steps, while the privileged expert π_e operates at the single-action level. At each DAgger iteration, we collect student rollouts and query the expert at student-visited inference states, executing it closed-loop for H steps to construct action-chunk labels. The resulting examples are aggregated into a replay buffer $\mathcal{D}_{\text{DAgger}}$, from which standard DAgger samples uniformly:

$$\mathcal{L}_{\text{DAgger}}(\theta) = \mathbb{E}_{i \sim \mathcal{D}_{\text{DAgger}}} \left[\ell_{\text{SFT}}(\pi_\theta(o_i), \mathbf{a}_i^e) \right]. \tag{1}$$

From imitation error to regret. While DAgger addresses covariate shift, its supervised objective does not distinguish action errors by their downstream consequences. The performance difference lemma (PDL) provides a complementary perspective, expressing the student–expert return gap in terms of expert-relative regret:

$$J(\pi_e) - J(\pi_\theta) = \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi_\theta},\, a\sim\pi_\theta}\left[R_e(s,a)\right], \quad R_e(s,a) = V_e(s) - Q_e(s,a), \tag{2}$$

where d^{π_θ} is the normalized discounted state distribution of the student. Thus, the contribution of a student action to the performance gap depends on its regret under expert continuation, rather than its disagreement with the expert in action space.

Directly optimizing this value-based objective is challenging for generative action-chunked policies due to the iterative denoising process and inability to compute $\pi(a|s)$. We instead retain DAgger's supervised learning objective and use the expert's critic to prioritize corrections to student actions with high estimated regret. The expert actor provides the supervision target, while its critic determines where to allocate learning effort under a fixed training budget.

Regret-weighted DAgger. For a student action chunk $\mathbf{a}_t$ executed for K steps, we estimate its return under expert continuation using the privileged critic:

$$\hat{Q}_e^{(K)}(s_t, \mathbf{a}_t) = \sum_{k=0}^{K-1} \gamma^k r_{t+k} + \gamma^K \hat{V}_e(s_{t+K}), \quad \hat{V}_e(s) = Q_e(s, \pi_e(s)). \tag{3}$$

The estimated regret is $\hat{R}_t = \hat{V}_e(s_t) - \hat{Q}_e^{(K)}(s_t, \mathbf{a}_t)$.

We prioritize examples in the aggregated DAgger buffer through regret-weighted sampling:

$$w_i = 1 + \lambda \,\mathrm{clip}\left(\frac{\hat{R}_i}{\sigma}, 0, c\right), \qquad p_i = \frac{w_i}{\sum_j w_j}, \tag{4}$$

where σ is the standard deviation of regret estimates within the relabeling batch, and λ and c control the strength and maximum extent of prioritization, respectively.

We sample with replacement according to p_i and optimize the standard supervised objective:

$$\mathcal{L}_{\mathrm{RW}}(\theta) = \mathbb{E}_{i\sim p}\left[\ell_{\mathrm{SFT}}(\pi_\theta(o_i), \mathbf{a}_i^e)\right]. \tag{5}$$

This preserves the native training objective of the generative student while allocating more updates to consequential mistakes, without requiring action log-probabilities or gradients through the critic. The additive baseline ensures nonzero sampling probability for all examples, while clipping bounds the relative sampling probabilities. We use $\lambda = 2$ and $c = 5$.

Our approach is related to cost-to-go-based interactive imitation learning [**? ?**]. Rather than directly optimizing expert cost-to-go, we use the regret of student actions to prioritize imitation of expert corrections.

3 Core Experiments

We present experiment results on the effectiveness of our insights on two long-horizon sparse reward tasks: (1) bimanual pick and **Hang Mug** and (2) single-arm **Leg Screwing** (cite). Both tasks are on a bimanual Franka embodiment with two-finger grippers. The pretrained policy we employ is $\pi_{0.5}$ finetuned on LBM data (cite) with a chunk-relative end-effector action space. In the case of Leg Screwing, since LBM data doesn't include this task, we study the expert learning with scripted policy demonstrations as our prior. We evaluate how to most effectively use the priors from the pretrained policy during expert learning. For distillation, we picked the hardest task from all experts learning results, based on difficulty to learn the task, to run experiments due to compute limitations.

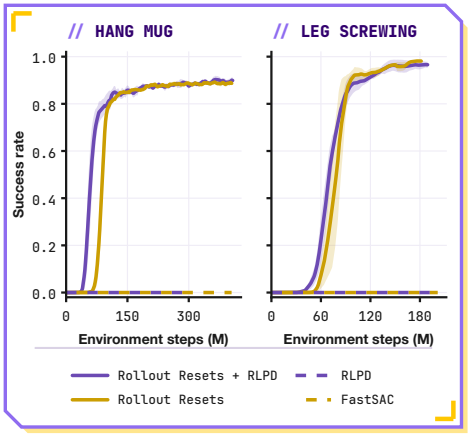


Figure 1: Success rate plots from initial conditions across the learning process for two tasks. Both tasks illustrate rollout resets (solid lines) is a necessary prior for successful expert learning and that RLPD can complement resets to speed up learning (solid purple).

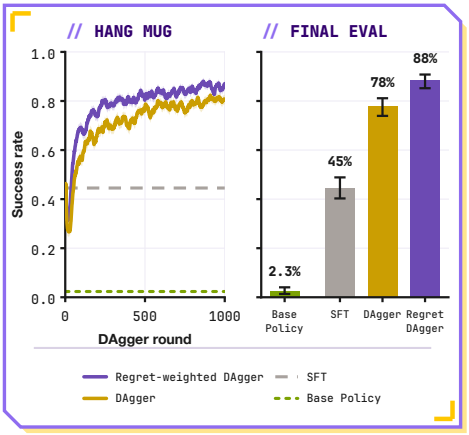


Figure 2: Regret-weighted DAgger outperforms DAgger in simulation for a fixed compute budget with statistical significance based on 512 evals per method.

3.1 Expert Learning with Guided Exploration

We investigate how pretrained behavior can alleviate the exploration burden of sparse-reward RL through two forms of guidance: an *action prior* and a *state-visitation prior*. We compare RL from task initial conditions, RLPD with offline policy-rollout transitions, resets sampled uniformly from policy-visited states, and the combination of both priors. In RLPD, each training batch has a ρ fraction of offline rollout transitions; the state prior is implemented by uniformly resetting to rollout-visited states.

As shown in Figure 1, initial-condition baselines (FastSAC and RLPD alone) remain near zero success on both Hang Mug and Leg Screwing, whereas rollout-state resets enable learning to high success. Adding RLPD can accelerates learning but doesn't make a step change in learning. These results suggest that rollout-derived reset states are a useful way to aid sparse-reward expert learning without manual engineering, while the action prior provides task-dependent complementary benefit.

3.2 Distillation Method Comparison

We compare standard DAgger and regret-weighted DAgger against the starting policy on Hang Mug at a fixed compute budget. As shown in Figure 2, both methods improve over DAgger rounds, with regret-weighted DAgger generally learning faster and finishing higher within the budget. After SFT warmup on expert demonstrations, the policy achieves 45% success; final success is 78% with DAgger and 88% with regret-weighted DAgger, across 512 evaluations per method. Thus, DAgger adds 33 percentage points over the starting policy, while regret-weighted DAgger adds 43 points overall and 10 points over standard DAgger. The figure reports the latter difference as statistically significant. These results support using the expert critic to focus distillation on student errors with greater estimated impact on task performance.

4 Discussion and Ongoing Work

Main unresolved questions:

- the $> 98\%$ expert to 80–90% student distillation gap;

- the sim-to-real gap;
- what recovery behaviors already exist in the generalist and distilled student;
- how to train explicit recovery from failure and out-of-distribution states;
- how to use failed trajectories as curriculum reset states; and
- critic- or Q-guided and RL-based distillation.
- lessons from sim2real transfer

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